```
!pip install lifelines -q
```

```
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 409.1/409.1 kB 7.3 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 117.3/117.3 kB 10.2 MB/s eta 0:00:00
  Building wheel for autograd-gamma (setup.py) ... done
```

```
from sklearn.datasets import load_breast_cancer
import pandas as pd

data = load_breast_cancer()
df = pd.DataFrame(data.data, columns=data.feature_names)
df['target'] = data.target

print(f"Dataset loaded: {len(df)} patients, {len(df.columns)} columns")
df.head()
```

```
Dataset loaded: 569 patients, 31 columns
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33	184.60
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41	158.80
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53	152.50
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50	98.87
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67	152.20

5 rows × 31 columns

```
!pip install lifelines -q
```

```
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 409.1/409.1 kB 7.7 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 117.3/117.3 kB 10.6 MB/s eta 0:00:00
  Building wheel for autograd-gamma (setup.py) ... done
```

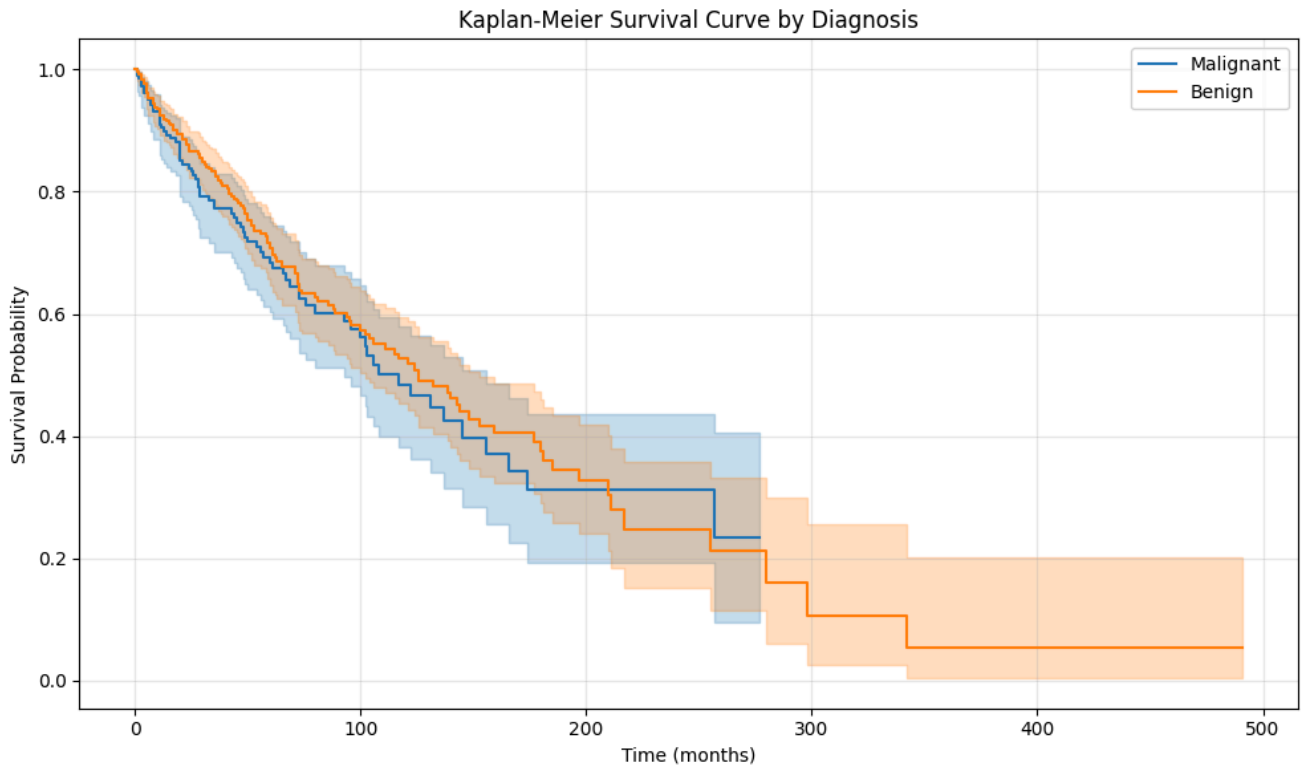
```
import numpy as np
from lifelines import KaplanMeierFitter
import matplotlib.pyplot as plt

# Add simulated survival data
np.random.seed(42)
df['duration'] = np.random.exponential(scale=60, size=len(df)).astype(int) + 1
df['event'] = np.random.binomial(1, 0.4, size=len(df))

# Split by diagnosis group using the 'target' column (0=malignant, 1=benign)
kmf = KaplanMeierFitter()

plt.figure(figsize=(10, 6))
for group, label in [(0, 'Malignant'), (1, 'Benign')]:
    mask = df['target'] == group
    kmf.fit(df['duration'][mask], event_observed=df['event'][mask], label=label)
    kmf.plot_survival_function(ci_show=True)

plt.title('Kaplan-Meier Survival Curve by Diagnosis')
plt.xlabel('Time (months)')
plt.ylabel('Survival Probability')
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```




```
from lifelines.statistics import logrank_test

malignant = df[df['target'] == 0]
benign = df[df['target'] == 1]

results = logrank_test(malignant['duration'], benign['duration'],
                       malignant['event'], benign['event'])
print(f"Log-rank test p-value: {results.p_value:.4f}")
print("Significant difference!" if results.p_value < 0.05 else "No significant difference.")
```

```
Log-rank test p-value: 0.5100
No significant difference.
```

```
from lifelines.datasets import load_rossi
from lifelines.datasets import load_gbsg2
import pandas as pd
from lifelines import KaplanMeierFitter
import matplotlib.pyplot as plt
from lifelines.statistics import logrank_test

# Load real breast cancer survival dataset (built into lifelines!)
df_real = load_gbsg2()
print(df_real.head())
print(df_real.columns.tolist())
```

```
  horTh  age menostat  tsize tgrade  pnodes  progrec  estrec  time  cens
0    no   70     Post     21     II       3       48      66  1814     1
1   yes   56     Post     12     II       7       61      77  2018     1
2   yes   58     Post     35     II       9       52     271   712     1
3   yes   59     Post     17     II       4       60      29  1807     1
4    no   73     Post     35     II       1       26      65   772     1
['horTh', 'age', 'menostat', 'tsize', 'tgrade', 'pnodes', 'progrec', 'estrec', 'time', 'cens']
```

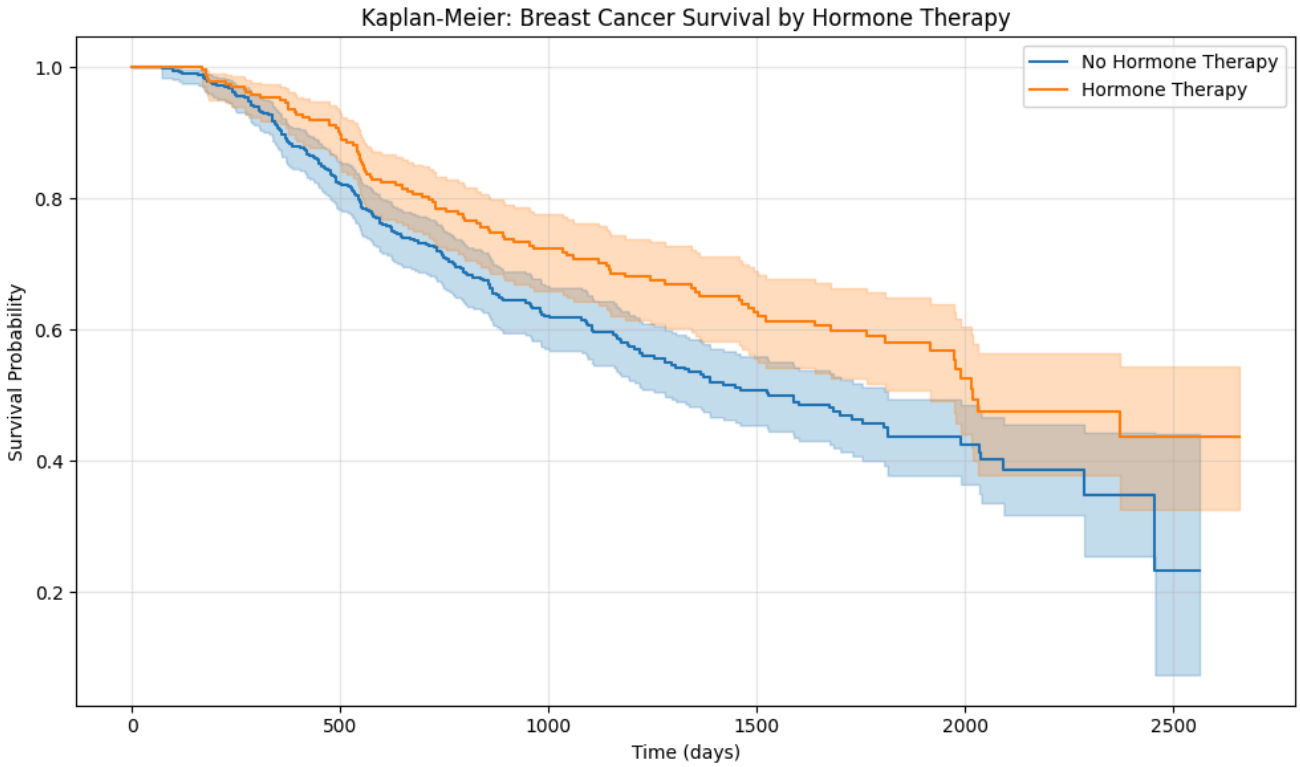
```
kmf = KaplanMeierFitter()
plt.figure(figsize=(10, 6))

for group, label in [('no', 'No Hormone Therapy'), ('yes', 'Hormone Therapy')]:
    mask = df_real['horTh'] == group
    kmf.fit(df_real['time'][mask], event_observed=df_real['cens'][mask], label=label)
    kmf.plot_survival_function(ci_show=True)

plt.title('Kaplan-Meier: Breast Cancer Survival by Hormone Therapy')
plt.xlabel('Time (days)')
plt.ylabel('Survival Probability')
```

```
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

# Log-rank test
g0 = df_real[df_real['horTh'] == 'no']
g1 = df_real[df_real['horTh'] == 'yes']
results = logrank_test(g0['time'], g1['time'], g0['cens'], g1['cens'])
print(f"Log-rank p-value: {results.p_value:.4f}")
print("Significant difference!" if results.p_value < 0.05 else "No significant difference.")
```



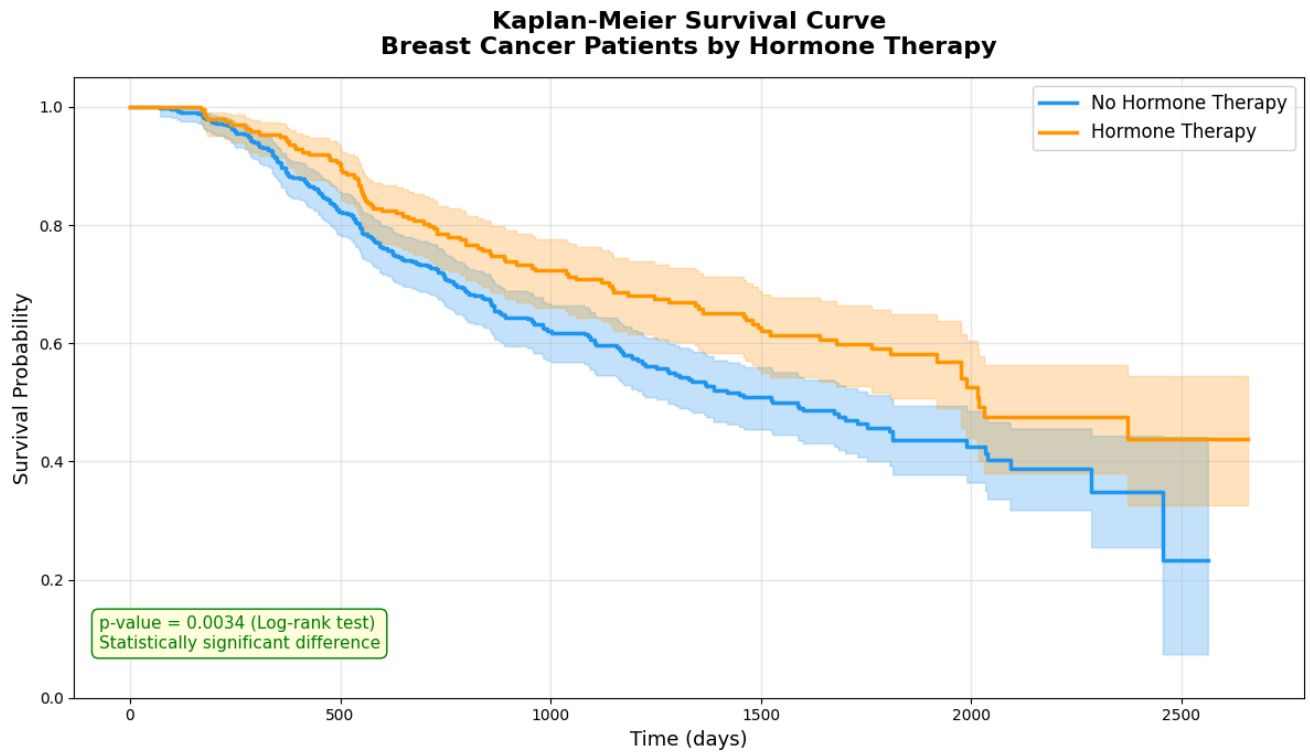

```
Log-rank p-value: 0.0034
Significant difference!
```

```
import matplotlib.pyplot as plt
from lifelines import KaplanMeierFitter

kmf = KaplanMeierFitter()
fig, ax = plt.subplots(figsize=(12, 7))

colors = ['#2196F3', '#FF9800']
for (group, label), color in zip([('no', 'No Hormone Therapy'), ('yes', 'Hormone Therapy')], colors):
    mask = df_real['horTh'] == group
    kmf.fit(df_real['time'][mask], event_observed=df_real['cens'][mask], label=label)
    kmf.plot_survival_function(ax=ax, ci_show=True, color=color, linewidth=2.5)

ax.set_title('Kaplan-Meier Survival Curve\nBreast Cancer Patients by Hormone Therapy',
             fontsize=16, fontweight='bold', pad=15)
ax.set_xlabel('Time (days)', fontsize=13)
ax.set_ylabel('Survival Probability', fontsize=13)
ax.legend(fontsize=12, loc='upper right')
ax.annotate('p-value = 0.0034 (Log-rank test)\nStatistically significant difference',
            xy=(0.02, 0.08), xycoords='axes fraction',
            fontsize=11, color='green',
            bbox=dict(boxstyle='round,pad=0.4', facecolor='lightyellow', edgecolor='green'))
ax.grid(alpha=0.3)
ax.set_ylim(0, 1.05)
plt.tight_layout()
plt.savefig('km_curve_polished.png', dpi=150, bbox_inches='tight')
plt.show()
print("Plot saved as km_curve_polished.png ✅")
```




```
Plot saved as km_curve_polished.png ✅
```

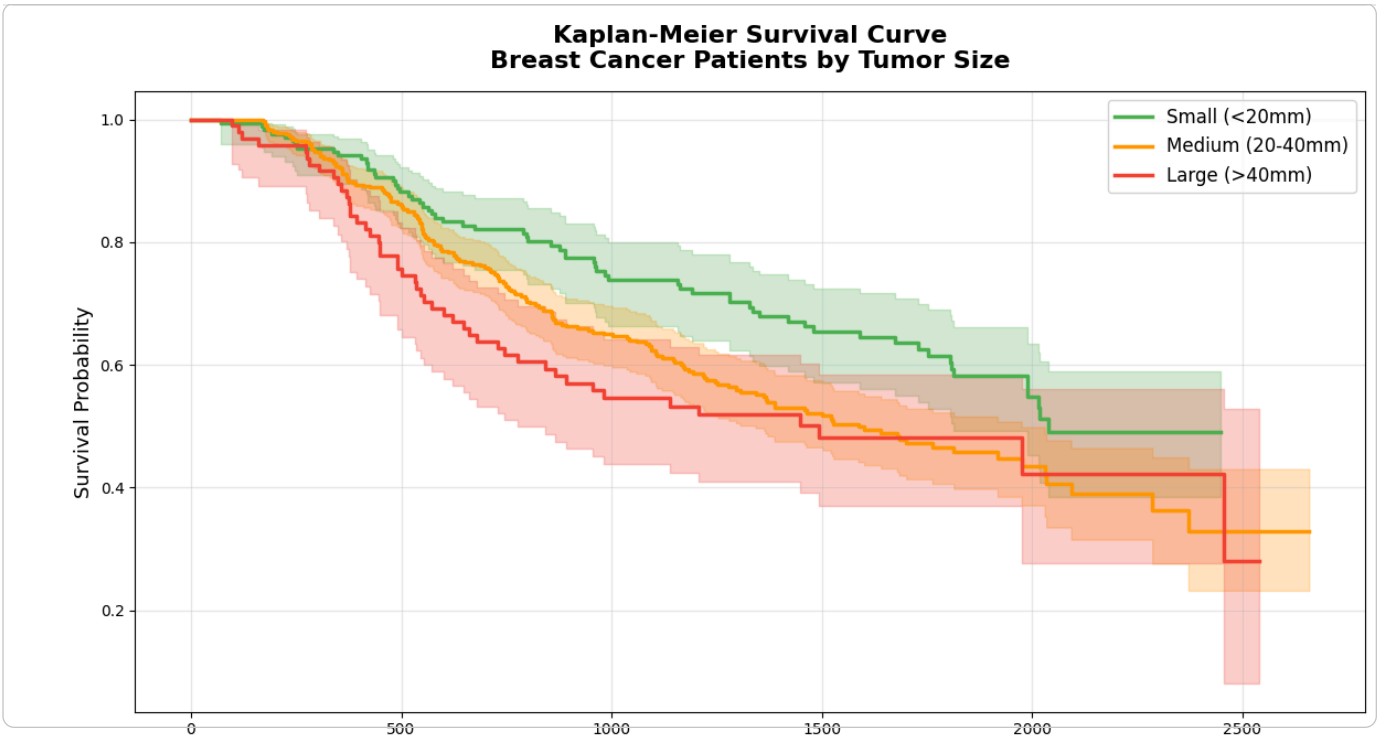
```
# Create tumor size groups
df_real['size_group'] = pd.cut(df_real['tsize'], bins=[0, 20, 40, 200],
                               labels=['Small (<20mm)', 'Medium (20-40mm)', 'Large (>40mm)'])

kmf = KaplanMeierFitter()
fig, ax = plt.subplots(figsize=(12, 7))

colors2 = ['#4CAF50', '#FF9800', '#F44336']
for (group, color) in zip(['Small (<20mm)', 'Medium (20-40mm)', 'Large (>40mm)'], colors2):
    mask = df_real['size_group'] == group
    kmf.fit(df_real['time'][mask], event_observed=df_real['cens'][mask], label=group)
    kmf.plot_survival_function(ax=ax, ci_show=True, color=color, linewidth=2.5)

ax.set_title('Kaplan-Meier Survival Curve\nBreast Cancer Patients by Tumor Size',
             fontsize=16, fontweight='bold', pad=15)
ax.set_xlabel('Time (days)', fontsize=13)
ax.set_ylabel('Survival Probability', fontsize=13)
ax.legend(fontsize=12)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()

# Log-rank test for tumor size
from lifelines.statistics import multivariate_logrank_test
results2 = multivariate_logrank_test(df_real['time'], df_real['size_group'], df_real['cens'])
print(f"Tumor size log-rank p-value: {results2.p_value:.4f}")
print("Significant difference!" if results2.p_value < 0.05 else "No significant difference.")
```



Start coding or generate with AI.

```
Significant difference!
```

📊 Analysis Summary

Dataset

- **Source:** GBSG2 — German Breast Cancer Study Group
- **Patients:** 686 women with primary node-positive breast cancer
- **Follow-up period:** Up to ~7 years (2,600+ days)

Key Findings

1. Hormone Therapy vs No Hormone Therapy

- Patients receiving hormone therapy showed consistently higher survival probability throughout the follow-up period.
- At day 2500: ~44% survival (hormone therapy) vs ~23% (no hormone therapy)
- Log-rank p-value = 0.0034 ✅ Statistically significant

2. Survival by Tumor Size

- Smaller tumors (<20mm) are associated with better long-term survival.
- Larger tumors (>40mm) show steeper early decline in survival probability.
- This confirms tumor size as an important prognostic factor.

Conclusion

Kaplan-Meier analysis successfully identified two significant predictors of breast cancer survival: hormone therapy treatment and tumor size. These findings are consistent with established clinical literature.